JOSHUA COWELL

Active Top Secret / SCI Clearance | **Full-Scope Polygraph**

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EDUCATION

Bachelor of Science in Mechanical Engineering Expected May 2027

Liberty University, Lynchburg, VA GPA: 3.85

Relevant Coursework: Fluid Dynamics, Heat Transfer, Thermodynamics I & II, Thermal-Fluids Laboratory, Differential Equations, Dynamics, Materials Engineering, Statics, Calculus I–III, Computer-Aided Engineering

Activities: American Nuclear Society (ANS) – Founding Member (Nov 2025); AIAA – Founding Member (Jul 2026)

Leadership: Liberty Rocketry – Chairman, Board of Advisors (Aug 2026 – Present)

PROFESSIONAL EXPERIENCE

Advanced Weapons Systems Analyst May 2026 – Present

Central Intelligence Agency (CIA)

- Conduct testing of warhead systems and analyze results to calibrate simulation models
- Develop a Python-based geospatial analysis tool for operational planning, leveraging Claude Mythos to refine the program and meet team requirements
- Conduct technical assessments of advanced unmanned systems and collaborate with multidisciplinary engineering teams to evaluate classified capabilities, translating findings into actionable intelligence for senior leadership and policymakers
- Brief high-level Agency officials and write concise analytic products that communicate complex technical data in clear, decision-ready formats

Thermal Hydraulics Engineering Intern May 2025 – Dec 2025

Framatome Nuclear

- Developed a predictive wear model in Python for safety-critical reactor components, achieving 75% improvement in accuracy over legacy methods
- Engineered a Python algorithm processing 6-dimensional datasets to calculate safe operating limits, reducing analysis time from 800+ hours to minutes
- Performed system-level thermal-hydraulic and flow-induced vibration (FIV) analysis to support component qualification
- Authored technical documentation and presented findings to engineering leadership; awarded $2,500 for presentation excellence

Chief Systems Engineer and Team Lead June 2025 – Aug 2026

Liberty Rocketry (Ranked 18th of 141 international teams at 2026 IREC)

- Led the design, testing, and analysis of high-powered rocket systems as member captain
- Oversaw engineering decisions across propulsion, avionics, recovery, and aerodynamics subsystems
- Facilitated communication among internal teams, external vendors, and university faculty to improve project execution
- Demonstrated advanced knowledge of SolidWorks, ANSYS, MATLAB, OpenRocket, and RocketPy to validate team members' models, simulations, and design work

Lead Undergraduate Research Assistant Sep 2024 – Present

Liberty University

- Lead a team of 5 researchers conducting split-Hopkinson pressure bar (SHPB) testing for dynamic material characterization
- Design test procedures and fixtures for high-strain-rate experiments; analyze results using electron microscopy and high-speed imaging

Camp Counselor May 2024 – Aug 2024

Woodlands Camp, GA

- Provided 24-hour supervision during weeklong outdoor expeditions with 12 youth (ages 8–18), instructing survival skills, rock climbing, whitewater rafting, and cave camping

TECHNICAL SKILLS

Analysis & CAE:	ANSYS Fluent, ANSYS Mechanical, AFT Fathom, OpenRocket, RocketPy, FEA, CFD
CAD & Design:	SolidWorks, Onshape, GD&T, Drafting Practices, 3D Printing, CNC Machining
Programming:	Python (NumPy, Xarray, Pandas), MATLAB, Data Analysis, Automation
Laboratory:	Scanning Electron Microscope (SEM), High-Speed Imaging, Material Testing

PROJECTS

Heat Exchanger Thermal-Hydraulic Optimization

- Performed CFD analysis using ANSYS Fluent to optimize heat exchanger thermal performance and flow characteristics

Unmanned Aerial Vehicle (UAV) Design & Analysis

- Designed a modular UAV airframe with rapid-prototyping capability; validated aerodynamics through CFD simulation

3D Printed Model Rocket

- Engineered a fully 3D printed rocket achieving Mach 0.5+ and 3,000 ft altitude; performed CFD analysis for aerodynamic optimization